

“With A Goal of 100 RISC-V Tape-Outs, 1-TOPS Is Poised To Become The World’s Largest Student-led SoC Tape-Out Programme”

A new wave is building in India’s chip design scene, and 1-TOPS is giving students a rare chance to create real, tape-out-ready RISC-V SoCs.



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What is the 1-TOPS programme? Can you explain the motivation behind the VLSI Society of India’s launch of the 1-TOPS initiative?

The “1-TOPS: 1 Tape-Out Per Student” programme, a unique collaboration between the VLSI Society of India (VSI) and C-DAC, is designed to empower B.Tech and M.Tech students with invaluable industry-ready experience. Through this initiative, students will complete a full chip System-on-Chip (SoC) Tape-Out before graduation.

With an ambitious goal of 100 RISC-V Tape-Outs, 1-TOPS is poised to become the world’s largest student-led SoC Tape-Out programme. Its core mission is to bridge the crucial gap between academia and industry, fostering a new generation of industry-aware and talented engineers for the chip design sector.

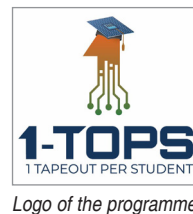
Under the 1-TOPS programme, industry experts led student teams will develop RISC-V cores and SoCs, managing the entire process from initial specifications to final Tape-Out, all within a one-year time-frame. These SoCs will be designed

and manufactured using advanced process nodes and industry-standard tools and methodologies.

The VLSI Society of India is very thankful to all semiconductor companies for their overwhelming support for the 1-TOPS programme.

Can you provide details on the programme and how students will benefit from it?

Under the 1-TOPS programme, students will gain hands-on experience by developing a RISC-V core from scratch and then integrating essential peripheral IPs to create a complete SoC. This initiative goes beyond traditional classroom learning for VLSI Design students, offering a unique opportunity to apply the fundamental concepts learned in their courses to a real-world chip design project and guiding them through every step of the process. Beyond technical expertise, participants will also develop crucial teamwork and soft professional skills vital for success



Logo of the programme

in the workplace, all while working collaboratively towards the ultimate goal of a Tape-Out.

Who can participate in this programme, and what background should participating students have?

The 1-TOPS programme welcomes B.Tech and M.Tech students from Electronics and Communication Engineering (ECE), Electronics and VLSI Design (EVD), or any other engineering or science discipline with an interest in VLSI Chip design. The launch of the B.Tech programme in Electronics and VLSI Design (EVD) by AICTE in more than 300 institutes has created massive interest among B.Tech students in the 1-TOPS programme.

Ideally, each 1-TOPS team should be a dynamic blend of students, ranging from second-year B.Tech onwards. While final-year B.Tech and M.Tech students are certainly encouraged to participate, teams must maintain a healthy mix of senior and junior members.